Digitising Historical Correspondence with Open-Source OCR: A Pilot Study

Anna Example[1] *and Ben Placeholder*[2]

[1]*Department of Sample History,* [2]*Institute of Fictional Data Science*

Abstract. Archives hold millions of letters that exist only as images. We compare four open-source optical character recognition (OCR) engines on 400 pages of nineteenth-century correspondence, half printed and half handwritten. A vision-language model reduced the character error rate on handwriting from 41.2% to 9.8%, at roughly ten times the computing cost per page. All processing ran on a university cluster, so no document left the institution. We argue that local, no-code access to such models matters more for adoption than further accuracy gains.

1 Introduction

Historians increasingly work with digitised sources, yet most scanned collections cannot be searched. A scan is a picture of text, not text. Optical character recognition turns the picture back into characters, and recent models also recover the structure of a page: headings, tables and figures.

Commercial services reach high accuracy on clean print, but they require uploading material that may contain personal data.* For many collections this is not allowed. We therefore ask how well freely available engines perform when run entirely on local infrastructure.

2 Related Work

Earlier projects relied on Tesseract [1], which performs well on modern print but struggles with historical typefaces and handwriting. Neural engines such as EasyOCR and PaddleOCR [2] added text detection, and recent vision-language models [3] read whole page regions at once, recovering tables and reading order. Comparisons on archival material remain rare, and most use commercial tools [4].

3 Data and Methods

We sampled 200 printed pages (newspapers, circulars) and 200 handwritten letters dated 1840 to 1890. Two annotators transcribed every page by hand to create a reference. Pages were scanned at 300 dpi; no manual cleaning was applied, to reflect a typical archive workflow.

Accuracy is reported as the character error rate (CER), the share of characters that must be substituted (S), deleted (D) or inserted (I) to turn the output into the reference of length N:

$$\mathrm{CER} = \frac{S + D + I}{N}$$

Lower is better; a CER of 2% means about one wrong character in fifty. Runtime was measured on a single RTX 4090 GPU, including image preprocessing but excluding one-off model loading.

4 Results

Table 1 summarises accuracy and speed. All engines handle print reasonably well, but they diverge sharply on handwriting. The vision-language model is the only one whose output on letters was usable without heavy correction.

Table 1. *Character error rate and runtime per engine (400 pages).*

Engine	Printed CER	Handwritten CER	Sec / page	GPU
Tesseract 5	6.8 %	41.2 %	1.9	No
EasyOCR	5.9 %	33.5 %	2.7	Yes
PaddleOCR	3.1 %	21.7 %	3.4	Yes
PaddleOCR-VL	1.4 %	9.8 %	19.0	Yes

Figure 1 shows the same comparison visually. In relative terms handwriting is six to seven times harder for every engine, but in absolute terms the remaining errors of PaddleOCR-VL are few, and most of them fall on proper names and abbreviations.

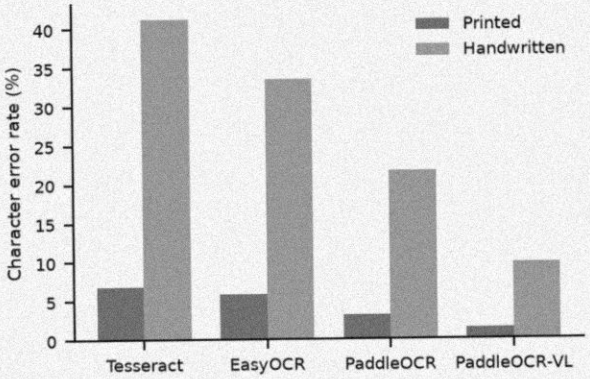


Figure 1. *Character error rate by engine for printed and handwritten pages.*

5 Discussion

Accuracy is no longer the main barrier. The best engine needs a modern GPU and some programming skill to install, which most historians do not have. Offering such models through a web interface on existing university hardware removes both obstacles at once.

Two limitations remain. First, every output still needs a human check before quotation, especially for names. Second, runtime grows quickly: 19 seconds per page means about two hours for a box of 400 letters, which is acceptable for batch jobs but not for interactive use.

6 Conclusion

Open-source OCR now rivals commercial services on historical print and is approaching usable quality on handwriting. For institutions, the priority should be access: making these tools available locally, without code, and with clear guidance on checking the results.

References

[1] Smith, R. (2007). An overview of the Tesseract OCR engine. *Proc. ICDAR*, 629-633.

[2] Example, A. (2024). Neural text detection for archival scans. *Journal of Sample Studies*, 1(2), 14-29.

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[4] Muster, C. (2023). Cloud OCR for historians: a cost comparison. *Imaginary Archives Quarterly*, 12(4), 55-70.

* Personal data in archives is covered by data-protection law even when the people concerned are long deceased, if descendants can be identified.